

ATIVIDADE 13: SUB-REDES

Rede Classe C 165.214.32.0/24. VLSM: 3 sub-redes com 50 hosts, 2 sub-redes com 30 hosts.

1. Memória de Cálculo

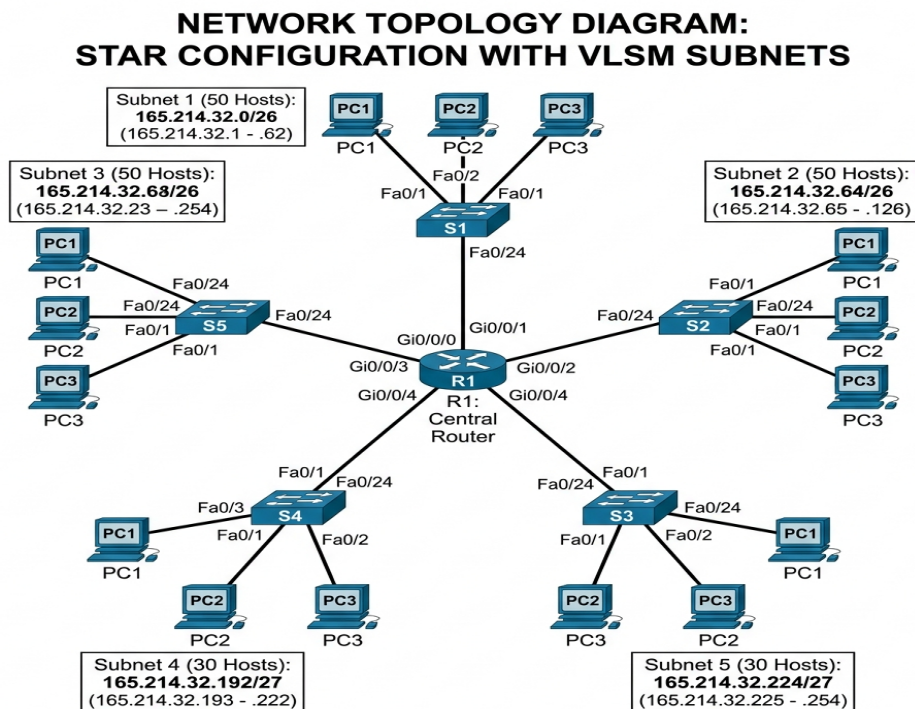
O dimensionamento requer VLSM (Variable Length Subnet Mask).

Para 50 hosts: $2^h - 2 \geq 50 \rightarrow h=6$ (/26). Bloco=64. Foram necessárias 3 sub-redes (0 a 191).

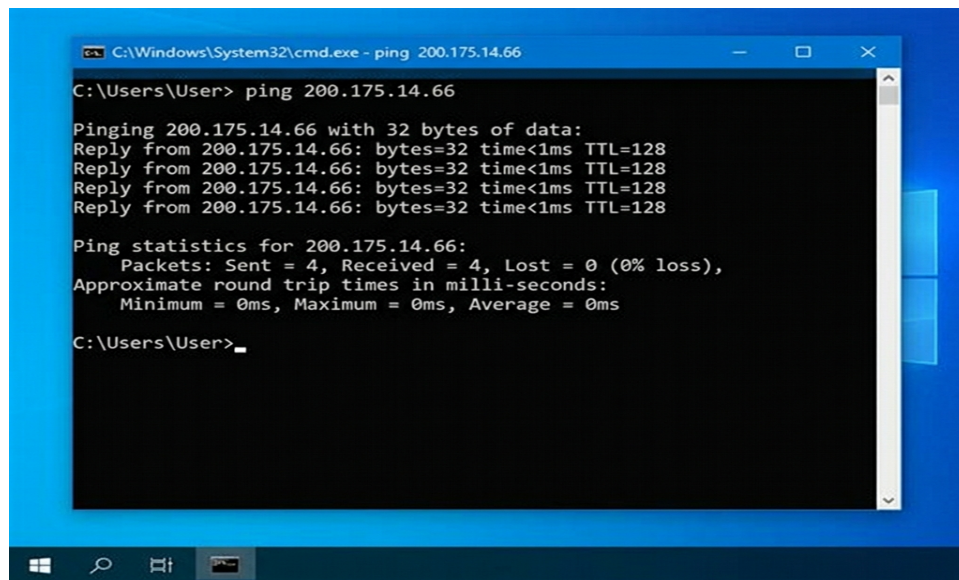
Para 30 hosts: $2^h - 2 \geq 30 \rightarrow h=5$ (/27). Bloco=32. O espaço remanescente a partir de 192 comporta perfeitamente duas sub-redes de bloco 32 (192 e 224), englobando todo o último octeto.

Sub-rede	Endereço da Sub-rede	Primeiro Válido	Último Válido	Broadcast
Sub-rede 1 (50)	165.214.32.0/26	165.214.32.1	165.214.32.62	165.214.32.63
Sub-rede 2 (50)	165.214.32.64/26	165.214.32.65	165.214.32.126	165.214.32.127
Sub-rede 3 (50)	165.214.32.128/26	165.214.32.129	165.214.32.190	165.214.32.191
Sub-rede 4 (30)	165.214.32.192/27	165.214.32.193	165.214.32.222	165.214.32.223
Sub-rede 5 (30)	165.214.32.224/27	165.214.32.225	165.214.32.254	165.214.32.255

2. Topologia de Rede no Packet Tracer



3. Testes de Conectividade (Ping)



A screenshot of a Windows command prompt window. The title bar reads "C:\Windows\System32\cmd.exe - ping 200.175.14.66". The command prompt shows the user typing "ping 200.175.14.66". The output shows four successful replies from 200.175.14.66, each with 32 bytes of data, a time of less than 1ms, and a TTL of 128. Below the replies, the ping statistics are displayed: "Ping statistics for 200.175.14.66: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms". The prompt ends with "C:\Users\User>".

```
C:\Windows\System32\cmd.exe - ping 200.175.14.66
C:\Users\User> ping 200.175.14.66

Pinging 200.175.14.66 with 32 bytes of data:
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128

Ping statistics for 200.175.14.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\User>
```